Cryptography

Diffie-Hellman Key Exchange and the Discrete Log Problem

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Diffie-Hellman Key Exchange

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Public-Key Cryptography



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Bruce Schneier



Twofish,
Blowfish

Vincent Rijmen
and Joan Daemen



AES

Modern private
key encryption

Rivest, Shamir
& Aldeman



Public-k
encrypt

Ron
Rivest



Hashing

Whitfield
Diffie



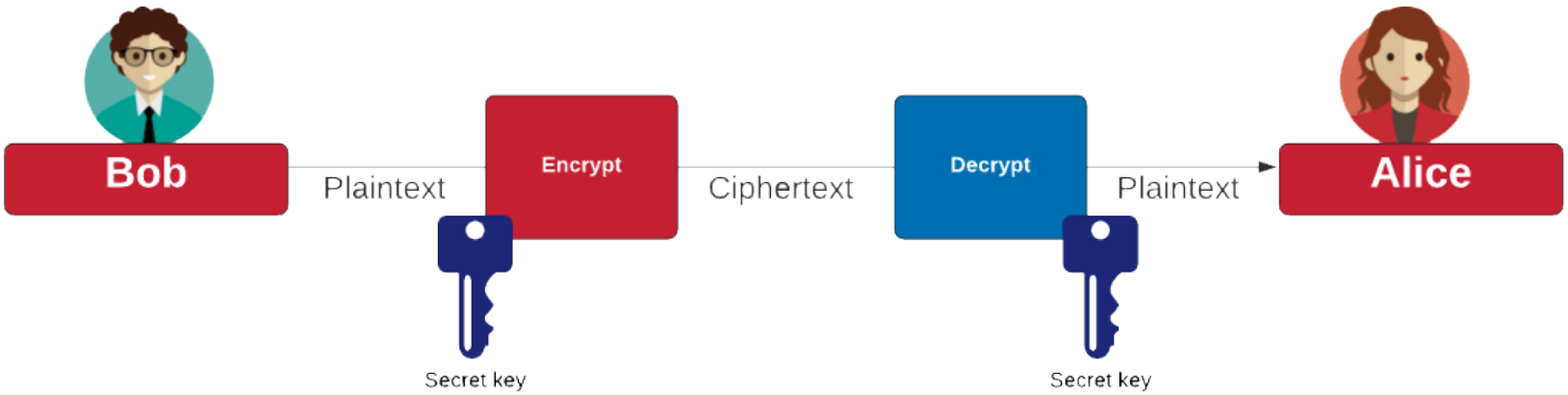
Key
interchange

Phil
Zimmerman



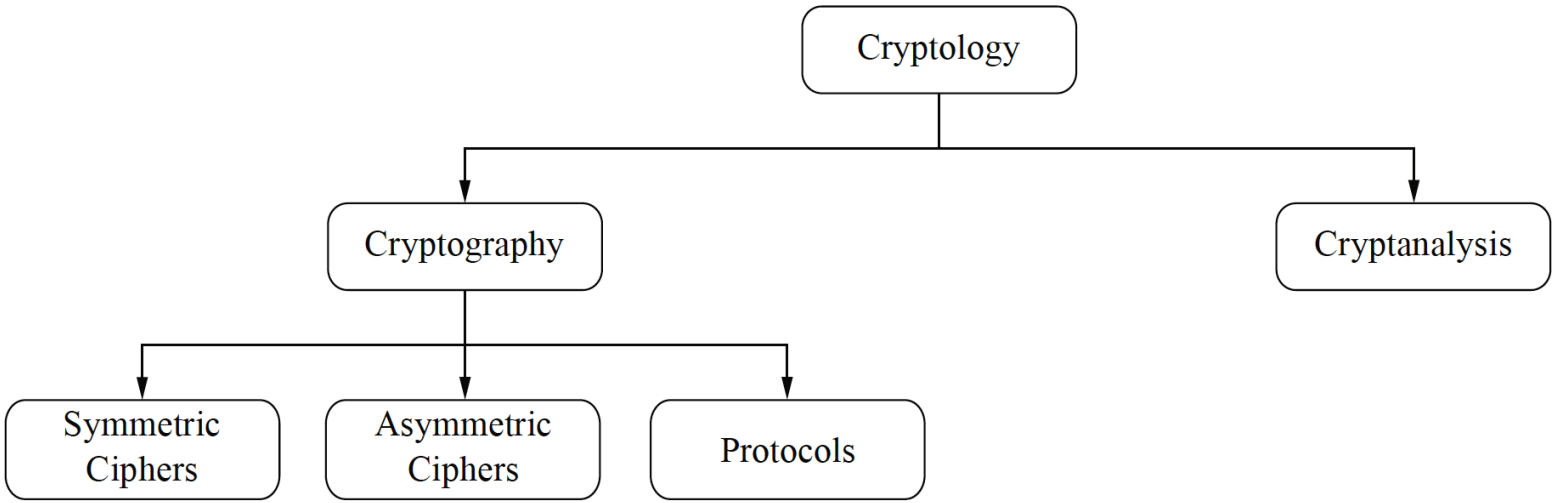
PGP
Encryption

Limitations of Symmetric Cryptography



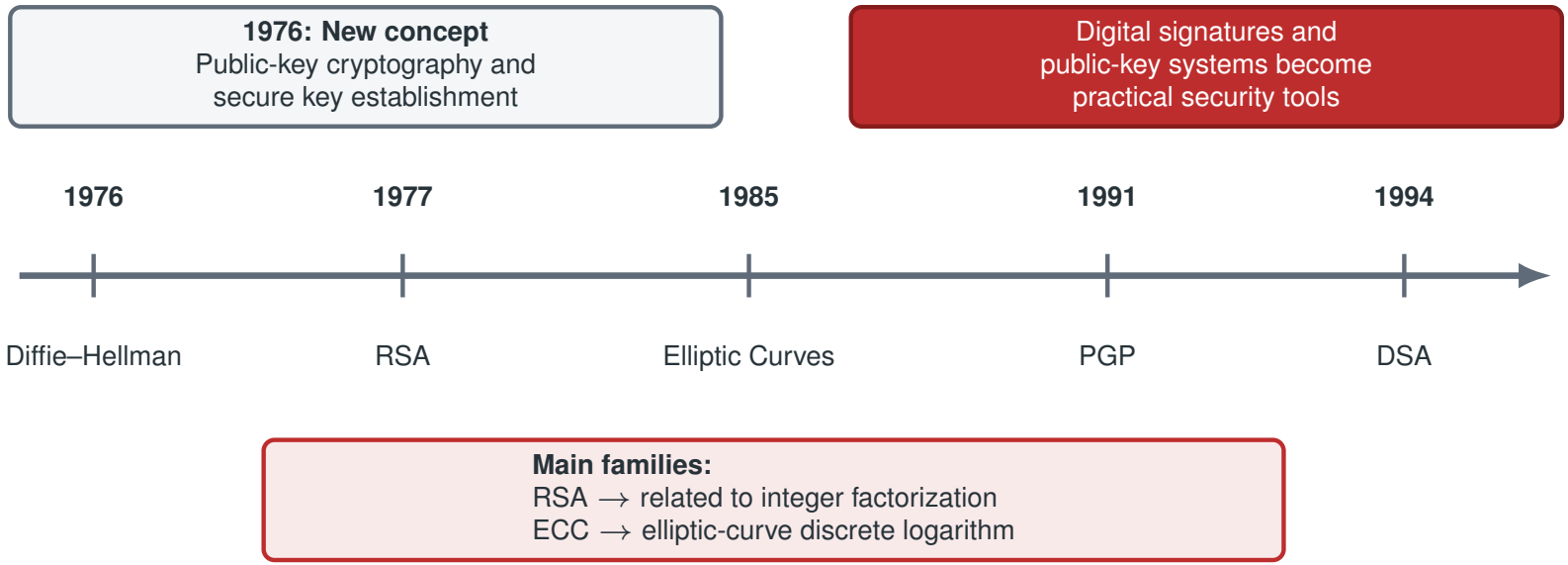
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Introduction to Public-Key Cryptography



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Evolution of Public-Key Cryptography



Historical Impact

Public-key cryptography enabled **key establishment, encryption, and digital signatures** without requiring a previously shared secret key. Public keys still need an authentication mechanism to prevent impersonation.

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Main Security Mechanisms of Public-Key Cryptography

1. Key Establishment

Create a shared secret over an insecure network.
Examples: Diffie–Hellman, RSA key transport.

2. Digital Signatures

Provide authentication and integrity, and support non-repudiation.
Examples: RSA, DSA, ECDSA.

3. Identification

Verify an entity with cryptographic challenge–response protocols.
Uses: smart cards, secure devices, authentication.

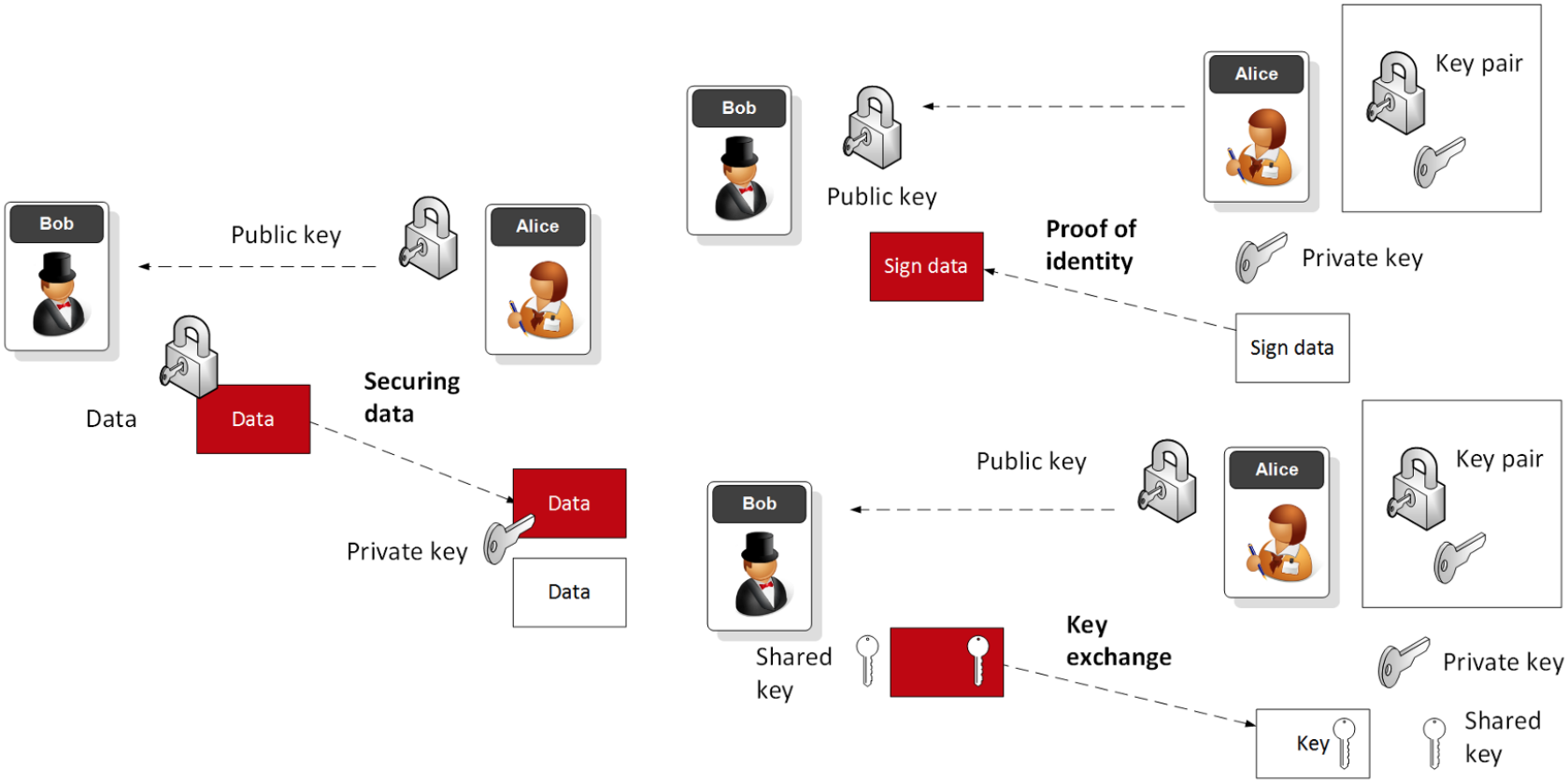
4. Encryption

Protect confidentiality using the recipient's public key.
Examples: RSA, ElGamal.

Public-key cryptography can establish security without a previously shared secret, but public keys must be authenticated.

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Public-Key Cryptography: Main Security Mechanisms



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Public-Key Algorithm Families

Public-Key Algorithm Families of Practical Relevance

Integer-Factorization Schemes Several public-key schemes are based on the fact that it is difficult to factor large integers. The most prominent representative of this algorithm family is RSA.

Discrete Logarithm Schemes There are several algorithms which are based on what is known as the discrete logarithm problem in finite fields. The most prominent examples include the Diffie–Hellman key exchange, Elgamal encryption or the Digital Signature Algorithm (DSA).

Elliptic Curve (EC) Schemes A generalization of the discrete logarithm algorithm are elliptic curve public-key schemes. The most popular examples include Elliptic Curve Diffie–Hellman key exchange (ECDH) and the Elliptic Curve Digital Signature Algorithm (ECDSA).

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Public-Key Algorithm Families

Integer Factorization	Discrete Logarithm	Elliptic Curves
Hard problem	**Hard problem**	**Hard problem**
Factor $n = pq$ for large primes p and q.	Given a generator g and $y = g^x \bmod p$, find x.	Given points P and $Q = kP$ in a large cyclic subgroup, find k.
Example	**Examples**	**Examples**
RSA	Diffie–Hellman, ElGamal, DSA	ECDH, ECDSA
Uses	**Uses**	**Uses**
Encryption, signatures	Key establishment, encryption, signatures	Key establishment, signatures

Each family relies on a computationally hard mathematical problem.

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Public-Key Cryptography Based on Discrete Logarithms

Diffie–Hellman Key Exchange (DHKE)

- DHKE allows two parties to establish a shared secret over an insecure channel.
- Its security is related to hard problems in finite cyclic groups, especially the discrete logarithm problem.
- Diffie–Hellman is used in protocols such as *SSH, TLS, and IPsec*.

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Public-Key Cryptography Based on Discrete Logarithms

Diffie–Hellman Key Exchange (DHKE)

Alice and Bob agree on the public parameters p and g.

$$\text{Alice:} \quad A = g^x \bmod p$$

$$\text{Bob:} \quad B = g^y \bmod p$$

They exchange A and B and independently compute:

$$\boxed{K = B^x \equiv A^y \equiv g^{xy} \pmod{p}}$$

- x and y are private; p, g, A, and B can be public.
- Computing $g^x \bmod p$ is efficient, but recovering x from $g^x \bmod p$ is computationally hard for suitable parameters.
- Both parties obtain the same key because $g^{xy} = g^{yx}$.

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Public-Key Cryptography Based on Discrete Logarithms

Diffie–Hellman Key Exchange (DHKE)

- DHKE has public domain parameters and private random exponents selected by each participant.

Diffie–Hellman Set-up

1. Choose a large prime p.
2. Choose a generator g of the group, or of a large prime-order subgroup of $\mathbb{Z}_p^*$.
3. Publish the domain parameters p and g.

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Key Exchange with Public Key

choose a random secret exponent b

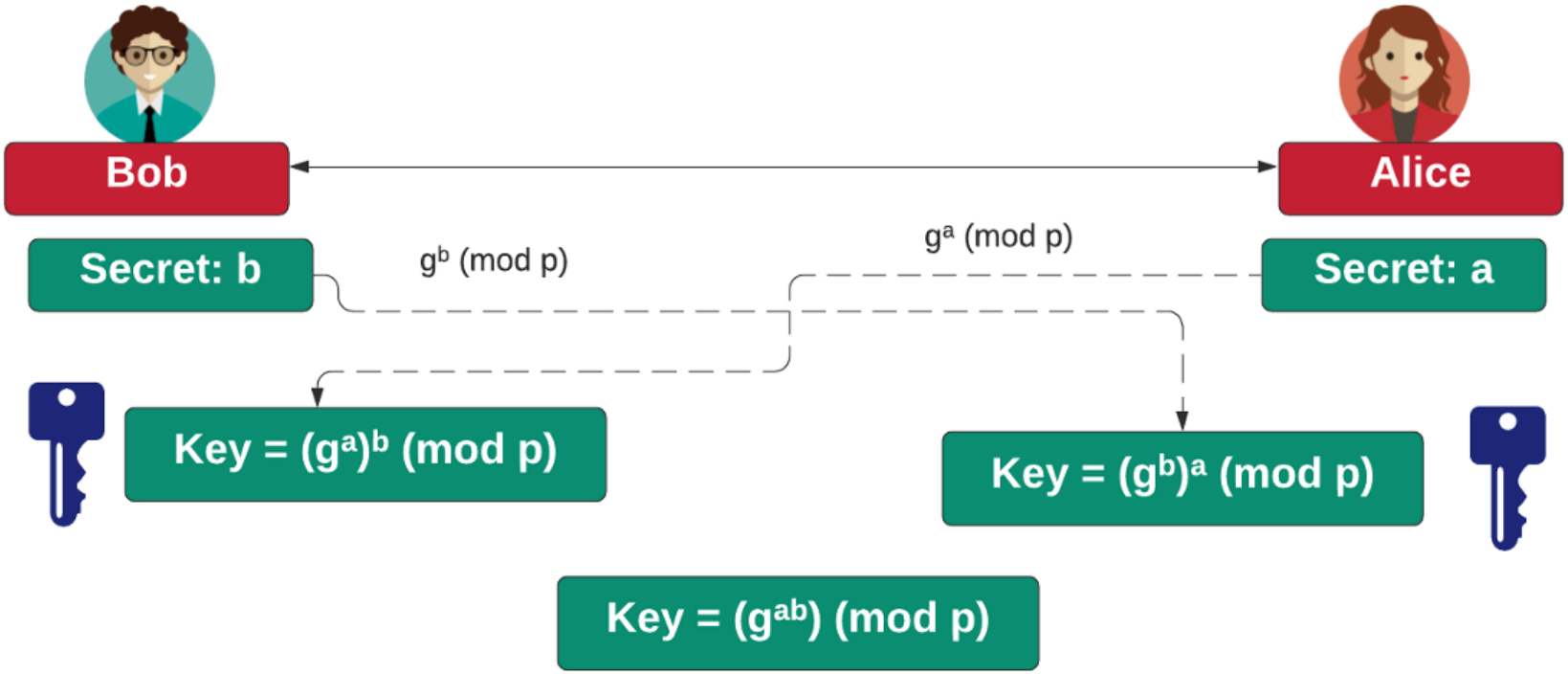
compute public value $B \equiv g^b \pmod{p}$

(Public)

(p**,** g**)**

choose a random secret exponent a

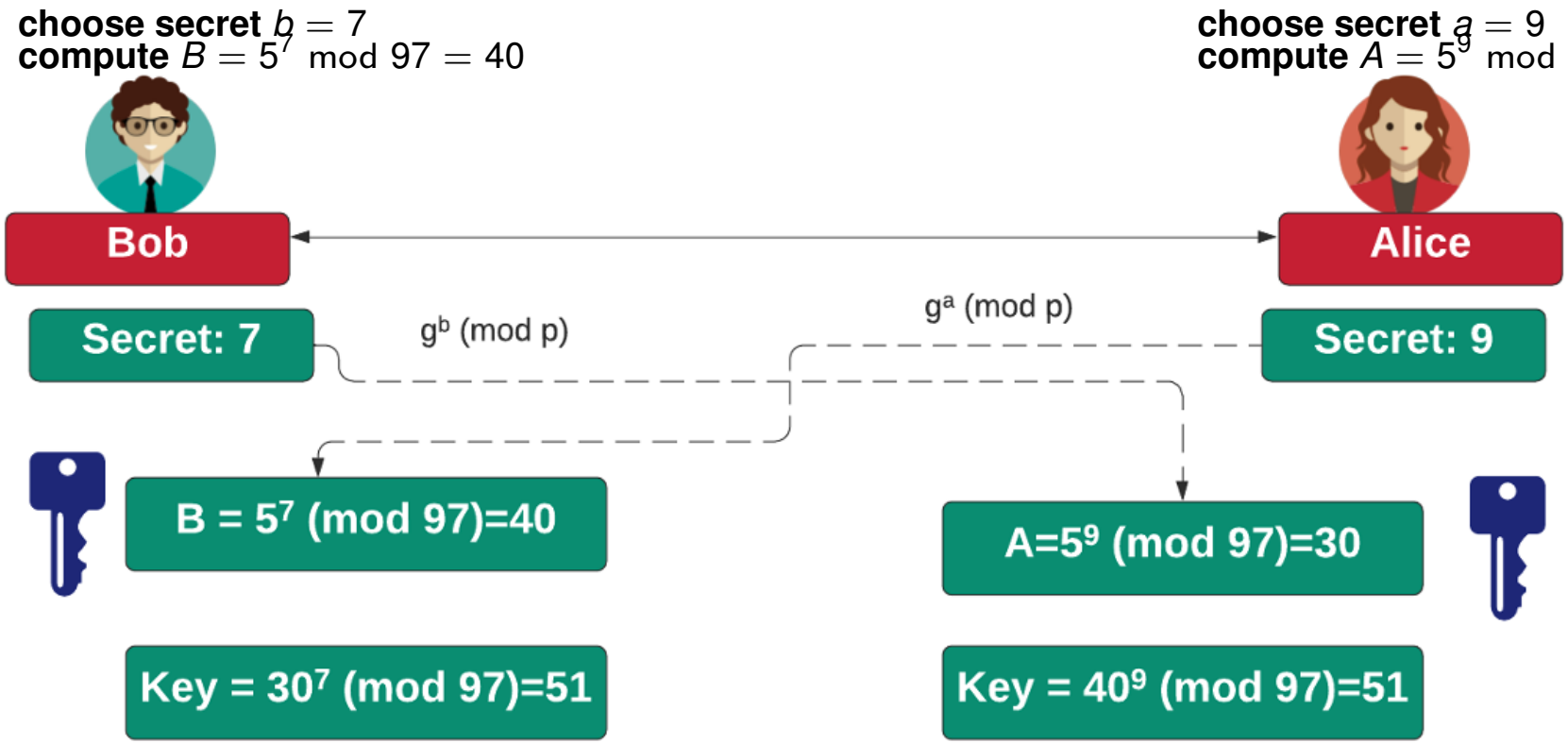
compute public value $A \equiv g^a \pmod{p}$



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Key Exchange with Public Key

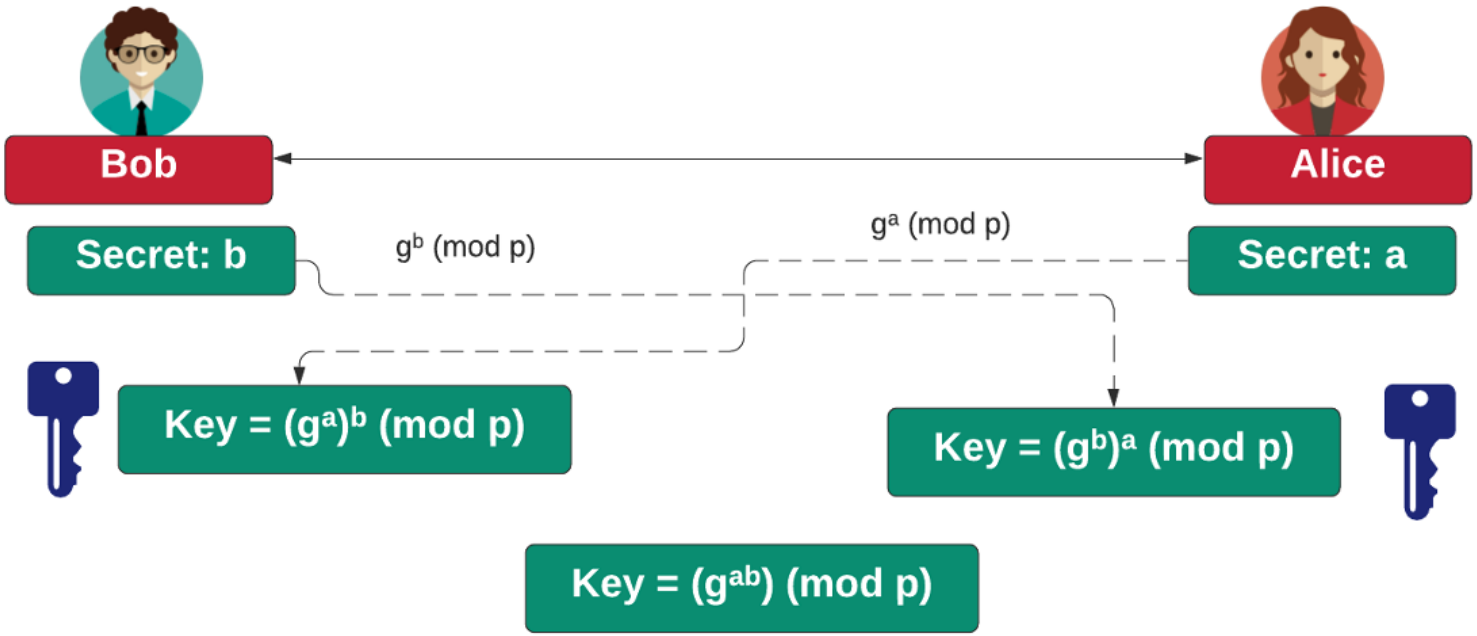
The Diffie-Hellman domain parameters are $p = 97$ and $g = 5$



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Key Exchange with Public Key

The Diffie–Hellman domain parameters are $p = 29$ and $g = 2$. Compute the shared secret K. In a real protocol, a key derivation function (KDF) is then used to derive the symmetric session key from K.



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Exercise: Diffie–Hellman Key Exchange

Alice and Bob want to establish a shared secret key over an insecure communication channel.

Public parameters:

$$p = 97, \qquad g = 5$$

Private keys:

$$a = 9 \text{ (Alice)}, \qquad b = 7 \text{ (Bob)}$$

Part I – Key Exchange

1. Compute the public keys of Alice and Bob.
2. Determine the shared secret key using both private keys.
3. Verify that both users obtain the same result.

Part II – Security Analysis

Eve intercepts the exchanged public keys and replaces both values with her own public key:

$$E = 5^{11} \bmod 97$$

4. Compute the new secret keys obtained by Alice and Bob.
5. Determine the two secret keys that Eve can calculate.
6. Explain why DHKE alone cannot prevent this attack.

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Exercise 1: Secure Messaging with DHKE

Alice and Bob want to establish a secure communication channel using Diffie–Hellman and AES-128.

Public parameters:

$$p = 467, \qquad g = 2$$

Private keys:

$$a = 37 \text{ (Alice)}, \qquad b = 53 \text{ (Bob)}$$

Tasks:

1. Compute the public keys A and B.
2. Calculate the shared secret independently for Alice and Bob.
3. Verify the mathematical relationship:

$$B^a \equiv A^b \pmod{p}$$

4. Describe how HKDF-SHA256 can be used to derive a 128-bit AES key from the shared secret.
5. Suppose an attacker replaces Alice's public key during transmission. Explain how this affects the shared secret computed by Bob.
6. Propose a mechanism to authenticate the exchanged public keys.

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Solution: Diffie–Hellman and AES

1. Public keys

$$A = g^a \bmod p = 2^{37} \bmod 467 = \boxed{263}$$

$$B = g^b \bmod p = 2^{53} \bmod 467 = \boxed{399}$$

2. Shared secret

Alice computes:

$$K_A = B^a \bmod p = 399^{37} \bmod 467 = \boxed{82}$$

Bob computes:

$$K_B = A^b \bmod p = 263^{53} \bmod 467 = \boxed{82}$$

3. Verification

$$B^a \equiv A^b \equiv g^{ab} \pmod{p}$$

4. AES-128 key derivation

The DH value should not be used directly as an AES key.

$$K_{\text{AES}} = \text{HKDF-SHA256}(\text{Encode}(82),\ \text{salt},\ \text{info},\ 16)$$

The output is a **16-byte (128-bit)** AES key.

5. Modified public key

If Eve replaces Bob's public key B with B_E, Alice computes

$$K'_A = B_E^a \bmod p$$

instead of $K = 82$.

- Alice and Bob no longer share the same key.
- Bob is unchanged if Alice's public key A is not modified.
- Replacing both public keys can create a Man-in-the-Middle attack.

6. Authentication

Authenticate the DH public values using **digital signatures** and **certificates**.

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Exercise 2: Key Exchange in an IoT Network

Three IoT sensors must establish independent secret keys with a central gateway before sending their data.

Public parameters:

$$p = 467, \qquad g = 2$$

The gateway uses the private key $b = 41$.
The sensors choose the following private keys:

$$a_1 = 17, \qquad a_2 = 29, \qquad a_3 = 43$$

Tasks:

1. Compute the gateway's public key B.
2. Determine the public key of each sensor:
$$A_i = g^{a_i} \bmod p$$
3. Calculate the three shared secrets:
$$K_i = B^{a_i} \bmod p$$
4. Verify each shared secret using the gateway's private key.
5. An attacker records the public key of Sensor 2 and sends it again in a later session.
If the gateway reuses $b = 41$, determine whether the resulting shared secret changes.
6. Explain how fresh private keys and authentication can reduce replay risks.

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Exercise 2: Solution – IoT Network

1. Gateway public key

$$B = 2^{41} \bmod 467 = \boxed{5}$$

2. Public keys and shared secrets

$$A_i = 2^{a_i} \bmod 467, \qquad K_i = A_i^{41} \bmod 467$$

Sensor	a_i	Public key A_i	Shared key K_i
1	17	312	350
2	29	240	226
3	43	20	446

3. Verification

$$K_i = B^{a_i} \bmod 467$$

Both the gateway and each sensor obtain the same key.

4. Replay attack
If Eve replays $A_2 = 240$ and the gateway reuses $b = 41$:

$$K_2 = 240^{41} \bmod 467 = \boxed{226}$$

Conclusion: The shared key remains unchanged.

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Exercise 2: Solution – IoT Network (cont.)

Protection: Use fresh private keys, authenticated session identifiers, and nonces to detect replay attacks.

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Exercise 3: Three-Party Diffie–Hellman

Alice, Bob, and Carol want to establish one shared secret key for a private group communication.

Public parameters:

$$p = 97, \qquad g = 5$$

Private keys:

$$a = 5, \qquad b = 7, \qquad c = 11$$

Tasks:

1. Compute the three public keys:
$$A = g^a, \quad B = g^b, \quad C = g^c \pmod{p}$$
2. Calculate the intermediate values:
$$X = B^a, \quad Y = C^b, \quad Z = A^c \pmod{p}$$
3. Alice sends X to Carol, Bob sends Y to Alice, and Carol sends Z to Bob. Calculate the final shared key for each user:
$$K_A = Y^a, \quad K_B = Z^b, \quad K_C = X^c \pmod{p}$$
4. Prove that all three users obtain:
$$K = g^{abc} \bmod p$$
5. If Carol leaves the group, explain why the remaining users should establish a new key.
6. Discuss whether this protocol provides authentication against an active attacker.

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Exercise 3: Solution – Three-Party DHKE

1. Public keys

$$A = 5^5 \bmod 97 = \boxed{21}$$

$$B = 5^7 \bmod 97 = \boxed{40}$$

$$C = 5^{11} \bmod 97 = \boxed{71}$$

2. Intermediate values

$$X = B^a = 40^5 \bmod 97 = \boxed{10}$$

$$Y = C^b = 71^7 \bmod 97 = \boxed{23}$$

$$Z = A^c = 21^{11} \bmod 97 = \boxed{57}$$

3. Final shared key

$$K_A = 23^5 \bmod 97 = \boxed{5}$$

$$K_B = 57^7 \bmod 97 = \boxed{5}$$

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Exercise 3: Solution – Three-Party DHKE (cont.)

$$K_C = 10^{11} \bmod 97 = \boxed{5}$$

4. Mathematical verification

$$K = g^{abc} = 5^{385} \bmod 97 = \boxed{5}$$

5. Security analysis

- When Carol leaves, establish a new group key.
- DHKE alone does not authenticate participants.
- Digital signatures can authenticate exchanged values.

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Finite cyclic groups

Groups



UC 2. Public-Key Cryptography and Key Establishment

- Number Theory for Public-Key Algorithms
- RSA
- Key Exchange
- Finite cyclic groups
- ElGamal public key encryption
- Elliptic Curves
- Using Private Key to Authenticate
- Public key encryption using a trapdoor function
- PGP: Signed Email

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Bruce Schneier



Twofish,
Blowfish

Vincent Rijmen
and Joan Daemen



AES

Modern private
key encryption

Rivest, Shamir
& Aldeman



Public-k
encrypt

Ron
Rivest



Hashing

Whitfield
Diffie



Key
interchange

Phil
Zimmerman



PGP
Encryption

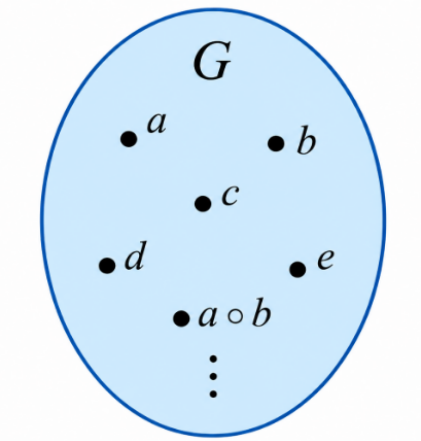
Groups

Fundamentals of abstract algebra

- **Groups**
- **Finite Groups**
- **Cyclic Groups**
- **Subgroups**

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Groups



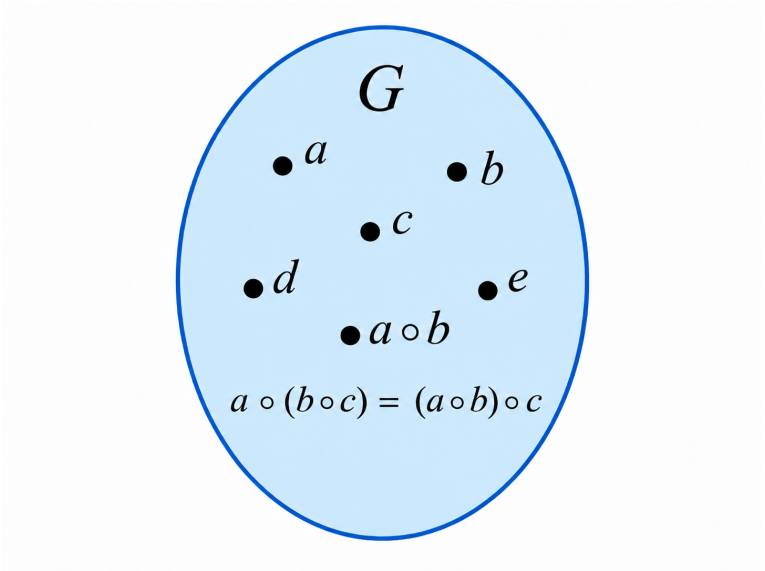
Definition

A group is a set of elements G together with an operation $\circ$ which combines two elements of G. A group has the following properties:

1. The group operation $\circ$ is closed. For all $a, b \in G$, the result $a \circ b = c \in G$.

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Groups

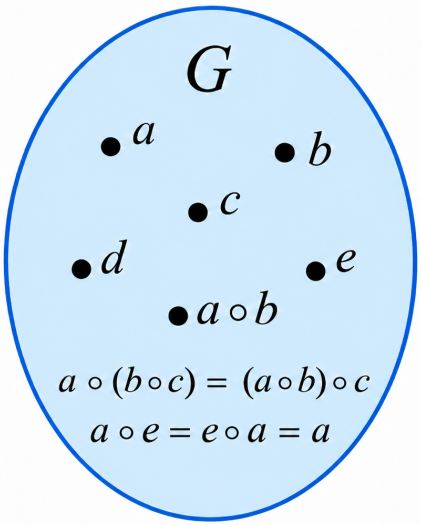


Properties of a Group

2. The group operation is associative. $a \circ (b \circ c) = (a \circ b) \circ c$ for all $a, b, c \in G$.

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Groups

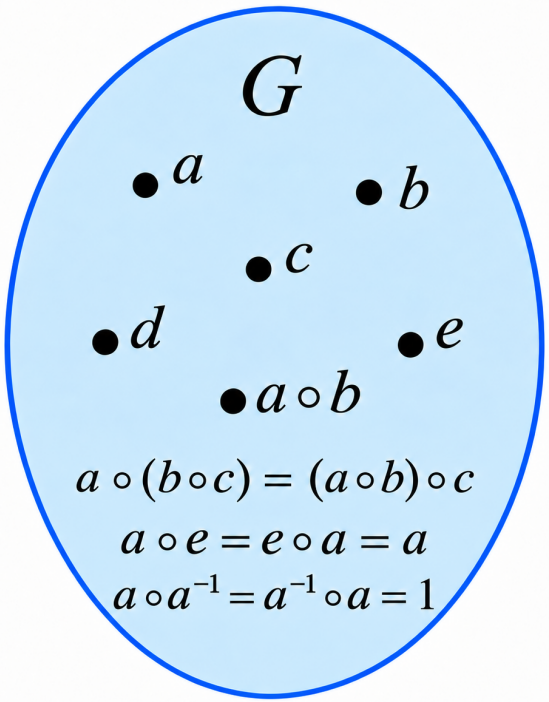


Properties of a Group

3. Identity element. There is an element $e \in G$, called the identity element, such that $a \circ e = e \circ a = a$ for all $a \in G$.

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Groups

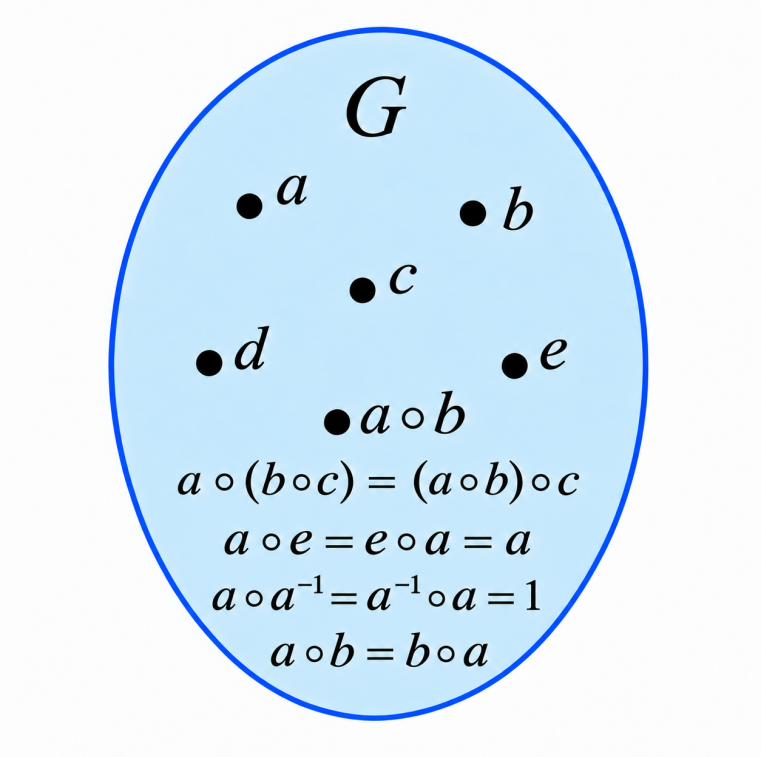


Properties of a Group

4. Inverse element. For each $a \in G$, there exists an element $a^{-1} \in G$ such that $a \circ a^{-1} = a^{-1} \circ a = e$.

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Groups



Properties of a Group

- A group G is abelian (or commutative) if, furthermore, $a \circ b = b \circ a$ for all $a, b \in G$

Additive Group of Integers $(\mathbb{Z}, +)$

$$\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$$

The operation is **integer addition**.

- **Closure:**
$$a + b \in \mathbb{Z} \qquad \forall\, a, b \in \mathbb{Z}$$
- **Associativity:**
$$a + (b + c) = (a + b) + c$$
- **Identity element:**
$$a + 0 = 0 + a = a$$
- **Inverse element:**
$$a + (-a) = (-a) + a = 0$$

Therefore, $(\mathbb{Z}, +)$ is a group.

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Multiplicative Group of Positive Real Numbers $(\mathbb{R}^+, \cdot)$

$$\mathbb{R}^+ = \{x \in \mathbb{R} \mid x > 0\}$$

The operation is **real number multiplication**.

- **Closure:**
$$a \cdot b \in \mathbb{R}^+ \qquad \forall\, a, b \in \mathbb{R}^+$$
- **Associativity:**
$$a \cdot (b \cdot c) = (a \cdot b) \cdot c$$
- **Identity element:**
$$a \cdot 1 = 1 \cdot a = a$$
- **Inverse element:**
$$a \cdot a^{-1} = a^{-1} \cdot a = 1, \qquad a^{-1} = \frac{1}{a}$$

Therefore, $(\mathbb{R}^+, \cdot)$ is a group.

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Exercise: Is $(\mathbb{Z}_9, \times)$ a Group?

Consider the set

$$\mathbb{Z}_9 = \{0, 1, 2, 3, 4, 5, 6, 7, 8\},$$

with the operation

$$a \circ b \equiv a \times b \pmod 9.$$

Question:
Is $(\mathbb{Z}_9, \circ)$ a group?

Verify the following group properties:

1. Closure
2. Associativity
3. Identity element
4. Inverse for every element

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Multiplicative Group of Integers Modulo n

The set $\mathbb{Z}_n^*$ contains the integers modulo n that are **coprime with** n:

$$\boxed{\mathbb{Z}_n^* = \{\, a \in \mathbb{Z}_n \mid \gcd(a, n) = 1 \,\}}$$

The operation is **multiplication modulo** n:

$$a \circ b \equiv a \cdot b \pmod{n}$$

Example: For $n = 9$,

$$\mathbb{Z}_9^* = \{1, 2, 4, 5, 7, 8\}$$

because each element is coprime with 9.

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Why is $(\mathbb{Z}_n^*, \cdot)$ a Group?

For all $a, b, c \in \mathbb{Z}_n^*$:

- **Closure:**

$$a \cdot b \pmod{n} \in \mathbb{Z}_n^*$$

- **Associativity:**

$$a \cdot (b \cdot c) \equiv (a \cdot b) \cdot c \pmod{n}$$

- **Identity element:**

$$a \cdot 1 \equiv 1 \cdot a \equiv a \pmod{n}$$

- **Inverse element:**

$$a \cdot a^{-1} \equiv a^{-1} \cdot a \equiv 1 \pmod{n}$$

Therefore, $(\mathbb{Z}_n^*, \cdot)$ is a multiplicative group.

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Groups

Elliptic Curve Groups $E(\mathbb{F}_p)$

- Set of Elements: These groups are defined over elliptic curves in finite fields, denoted as $E(\mathbb{F}_p)$, where $\mathbb{F}_p$ is a finite field of prime order p.
- Operation: The operation is point addition on the elliptic curve.
- Identity Element: The identity element is the point at infinity.
- Inverse Element: For each point on the curve, there exists a corresponding point that is its additive inverse.

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Elliptic Curve Group $E(\mathbb{F}_p)$

For a prime $p > 3$, consider the elliptic curve

$$E: \quad y^2 \equiv x^3 + Ax + B \pmod{p},$$

with

$$4A^3 + 27B^2 \not\equiv 0 \pmod{p}.$$

The group is

$$E(\mathbb{F}_p) = \{(x, y) \in \mathbb{F}_p^2 : y^2 \equiv x^3 + Ax + B \pmod{p}\} \cup \{\mathcal{O}\}.$$

- **Closure:**
$$P + Q \in E(\mathbb{F}_p)$$
- **Associativity:**
$$P + (Q + R) = (P + Q) + R$$
- **Identity element:**
$$P + \mathcal{O} = \mathcal{O} + P = P$$
- **Inverse element:**
$$P = (x, y) \quad \Rightarrow \quad -P = (x, -y \bmod p)$$
$$P + (-P) = \mathcal{O}$$

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Theorem: Multiplicative Group Modulo n

For $n \geq 2$, define

$$\boxed{\mathbb{Z}_n^* = \{\, a \in \{1, \ldots, n-1\} \mid \gcd(a, n) = 1 \,\}}$$

Under multiplication modulo n,

$$a \circ b \equiv a \cdot b \pmod{n},$$

the set $\mathbb{Z}_n^*$ forms an **abelian group**.

- **Identity:**
$$1 \in \mathbb{Z}_n^*, \qquad a \cdot 1 \equiv a \pmod{n}$$
- **Inverse:**
$$\forall\, a \in \mathbb{Z}_n^*, \quad \exists\, a^{-1} \in \mathbb{Z}_n^* : a\,a^{-1} \equiv 1 \pmod{n}$$
- **Commutativity:**
$$a \cdot b \equiv b \cdot a \pmod{n}$$

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Verify the Group Properties for $\mathbb{Z}_9^*$

If we choose $n = 9$, $\mathbb{Z}_n^*$ consists of the elements $\{1, 2, 4, 5, 7, 8\}$

$\times \bmod 9$	**1**	**2**	**4**	**5**	**7**	**8**
1	1	2	4	5	7	8
2	2	4	8	1	5	7
4	4	8	7	2	1	5
5	5	1	2	7	8	4
7	7	5	1	8	4	2
8	8	7	5	4	2	1

Multiplication table for $\mathbb{Z}_9^*$

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Finite Group

Definition

A group $(G, \circ)$ is called **finite** if it contains a finite number of elements. The number of elements in G is called the **order** or **cardinality** of the group:

$$\boxed{|G|}$$

Example: $(\mathbb{Z}_n^*, \cdot)$
The group

$$\mathbb{Z}_n^* = \{\, a \in \{1, \ldots, n-1\} \mid \gcd(a, n) = 1 \,\}$$

contains all integers modulo n that are coprime with n.
Its order is given by Euler's phi function:

$$\boxed{|\mathbb{Z}_n^*| = \phi(n)}$$

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Finite Group

Example: $(\mathbb{Z}_n^*, \cdot)$

For $n = 9$:

$$\mathbb{Z}_9^* = \{1, 2, 4, 5, 7, 8\}$$

and

$$|\mathbb{Z}_9^*| = \phi(9) = 9\left(1 - \frac{1}{3}\right) = 6.$$

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Finite Group

Order of an Element

Let a be an element of a finite group G. The **order** of an element $a \in G$, denoted by $\text{ord}(a)$, is the smallest positive integer k such that

$$\boxed{a^k = e}$$

where

$$a^k = \underbrace{a \circ a \circ \cdots \circ a}_{k \text{ times}}$$

and e is the identity element of the group.

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Finite Group (cont.)

Example: Order of 3 in $\mathbb{Z}_{11}^*$

We compute successive powers modulo 11:

$$3^1 \equiv 3 \pmod{11}$$
$$3^2 \equiv 9 \pmod{11}$$
$$3^3 \equiv 27 \equiv 5 \pmod{11}$$
$$3^4 \equiv 15 \equiv 4 \pmod{11}$$
$$3^5 \equiv 12 \equiv 1 \pmod{11}$$

Since 5 is the smallest positive exponent for which

$$3^5 \equiv 1 \pmod{11},$$

$$\boxed{\text{ord}(3) = 5}$$

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Cyclic Group

Cyclic Group

Definition

A finite group G is called **cyclic** if there exists an element $\alpha \in G$ such that

$$\boxed{\text{ord}(\alpha) = |G|}$$

Such an element α is called a **generator** of the group. In multiplicative groups, it is also often called a primitive element.

In multiplicative notation,

$$\boxed{G = \langle \alpha \rangle = \{\alpha^0, \alpha^1, \alpha^2, \ldots, \alpha^{|G|-1}\}}$$

Therefore, every element $a \in G$ can be written as

$$a = \alpha^i$$

for some integer i.

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Cyclic Group

- Check whether $a = 2$ is a generator of $\mathbb{Z}_{11}^*$.

Cyclic Group

- Check whether $a = 2$ is a generator of $\mathbb{Z}_{11}^* = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$.
- Cardinality of the group is $|\mathbb{Z}_{11}^*| = 10$.

$$a = 2 \qquad a^6 = 9 \bmod 11$$
$$a^2 = 4 \qquad a^7 = 7 \bmod 11$$
$$a^3 = 8 \qquad a^8 = 3 \bmod 11$$
$$a^4 = 5 \bmod 11 \qquad a^9 = 6 \bmod 11$$
$$a^5 = 10 \bmod 11 \qquad a^{10} = 1 \bmod 11$$
$$ord(a) = 10 = |\mathbb{Z}_{11}^*|$$

- This implies that (*i*) $a = 2$ is a generator and (*ii*) the group $\mathbb{Z}_{11}^*$ is cyclic.

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Exercise: Is 3 a Generator of $\mathbb{Z}_{11}^*$?

Consider the multiplicative group

$$\mathbb{Z}_{11}^* = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}.$$

Determine whether

$$\boxed{a = 3}$$

is a **generator** of $\mathbb{Z}_{11}^*$.

Compute the successive powers of 3 modulo 11:

$$3^i \pmod{11}, \qquad i = 1, \ldots, 10.$$

i	1	2	3	4	5	6	7	8	9	10
$3^i \bmod 11$										

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Exercise: Is 3 a Generator of $\mathbb{Z}_{11}^*$? (cont.)

Questions:

1. What is ord(3) in $\mathbb{Z}_{11}^*$?
2. Do the powers of 3 generate all elements of $\mathbb{Z}_{11}^*$?
3. Is 3 a generator of the group? Justify your answer.

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Properties of Cyclic Groups

Theorem 1

For every prime number p,

$$\boxed{\mathbb{Z}_p^*}$$

is a finite, abelian, and cyclic group under multiplication modulo p. Moreover,

$$\boxed{|\mathbb{Z}_p^*| = p - 1}$$

Therefore, there exists at least one generator α such that

$$\text{ord}(\alpha) = p - 1.$$

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Exercise: Verify that $\mathbb{Z}_7^*$ is Cyclic

Consider the multiplicative group

$$\mathbb{Z}_7^* = \{1, 2, 3, 4, 5, 6\}.$$

Use multiplication modulo 7.

Tasks:

1. Determine the order of the group:

$$|\mathbb{Z}_7^*| = ?$$

2. Verify that multiplication modulo 7 is commutative.
3. Compute the powers of 3 modulo 7:

$$3^1, 3^2, \ldots, 3^6 \pmod 7.$$

4. Determine $\mathrm{ord}(3)$.
5. Is 3 a generator of $\mathbb{Z}_7^*$?

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Exercise: Verify that $\mathbb{Z}_7^*$ is Cyclic

The group has

$$|\mathbb{Z}_7^*| = 6.$$

Multiplication modulo 7 is commutative:

$$ab \equiv ba \pmod 7.$$

Now compute the powers of 3:

i	1	2	3	4	5	6
$3^i \bmod 7$	3	2	6	4	5	1

Therefore,

$$\{3, 2, 6, 4, 5, 1\} = \mathbb{Z}_7^*.$$

Also,

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Exercise: Verify that $\mathbb{Z}_7^*$ is Cyclic (cont.)

$$3^6 \equiv 1 \pmod 7$$

and no smaller positive exponent gives 1. Thus,

$$\boxed{\text{ord}(3) = 6}$$

Since

$$\text{ord}(3) = |\mathbb{Z}_7^*|,$$

$$\boxed{3 \text{ is a generator of } \mathbb{Z}_7^*.}$$

Hence, $\mathbb{Z}_7^*$ is a finite, abelian, cyclic group.

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Properties of Cyclic Groups

Theorem 2: Element Order

Let G be a finite group and let $a \in G$. Then

$$\boxed{\text{ord}(a) \mid |G|}$$

That is, the order of every element divides the order of the group. As a consequence,

$$\boxed{a^{|G|} = e}$$

where e is the identity element of G.

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Exercise: Verify Theorem 2

Consider the group

$$\mathbb{Z}_{11}^* = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

and the element

$$a = 3.$$

Verify that:

$$\boxed{\text{ord}(3) \mid |\mathbb{Z}_{11}^*|}$$

and

$$\boxed{3^{|\mathbb{Z}_{11}^*|} \equiv 1 \pmod{11}}$$

Tasks:

1. Find $|\mathbb{Z}_{11}^*|$.
2. Compute ord(3).
3. Verify both statements of Theorem 2.

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Exercise: Verify Theorem 2

First,

$$|\mathbb{Z}_{11}^*| = 10.$$

Now compute the powers of 3:

i	1	2	3	4	5
$3^i \bmod 11$	3	9	5	4	1

Thus,

$$\boxed{\text{ord}(3) = 5}$$

and

$$\boxed{5 \mid 10}.$$

Therefore,

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Exercise: Verify Theorem 2 (cont.)

$$\text{ord}(3) \mid |\mathbb{Z}_{11}^*|.$$

Also,

$$3^{10} = (3^5)^2 \equiv 1^2 \equiv 1 \pmod{11}.$$

Hence,

$$\boxed{3^{|\mathbb{Z}_{11}^*|} \equiv 1 \pmod{11}}$$

Both properties of Theorem 2 are verified.

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Exercise: Orders of Elements in $\mathbb{Z}_{11}^*$

Consider the multiplicative group

$$\mathbb{Z}_{11}^* = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}.$$

Since

$$|\mathbb{Z}_{11}^*| = 10,$$

By Theorem 2, $\text{ord}(a) \mid 10$.

Tasks:

1. Determine the possible orders of the elements.
2. Find the order of each element in $\mathbb{Z}_{11}^*$.
3. Identify the generators of the group.

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Exercise: Orders of Elements in $\mathbb{Z}_{11}^*$

The divisors of 10 are

$$\boxed{1,\ 2,\ 5,\ 10}.$$

Therefore, these are the only possible orders.

Order	Elements
1	1
2	10
5	3, 4, 5, 9
10	2, 6, 7, 8

Thus,

$$\text{ord}(a) \mid 10 \qquad \forall\, a \in \mathbb{Z}_{11}^*.$$

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Exercise: Orders of Elements in $\mathbb{Z}_{11}^*$ (cont.)

The elements whose order is equal to the group order are

$$\boxed{2,\ 6,\ 7,\ 8}.$$

Therefore, these elements are the **generators** of $\mathbb{Z}_{11}^*$.

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Properties of Cyclic Groups

Theorem 3: Generators in a Finite Cyclic Group

Let G be a finite cyclic group.

1. The number of generators of G is

$$\boxed{\phi(|G|)}$$

   where ϕ is Euler's phi function.
2. If $|G|$ is prime, every non-identity element of G is a generator.

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Properties of Cyclic Groups

Example: $\mathbb{Z}_{11}^*$
Since

$$|\mathbb{Z}_{11}^*| = 10,$$

the number of generators is

$$\phi(10) = 10\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{5}\right) = 4.$$

Therefore, $\mathbb{Z}_{11}^*$ has exactly four generators:

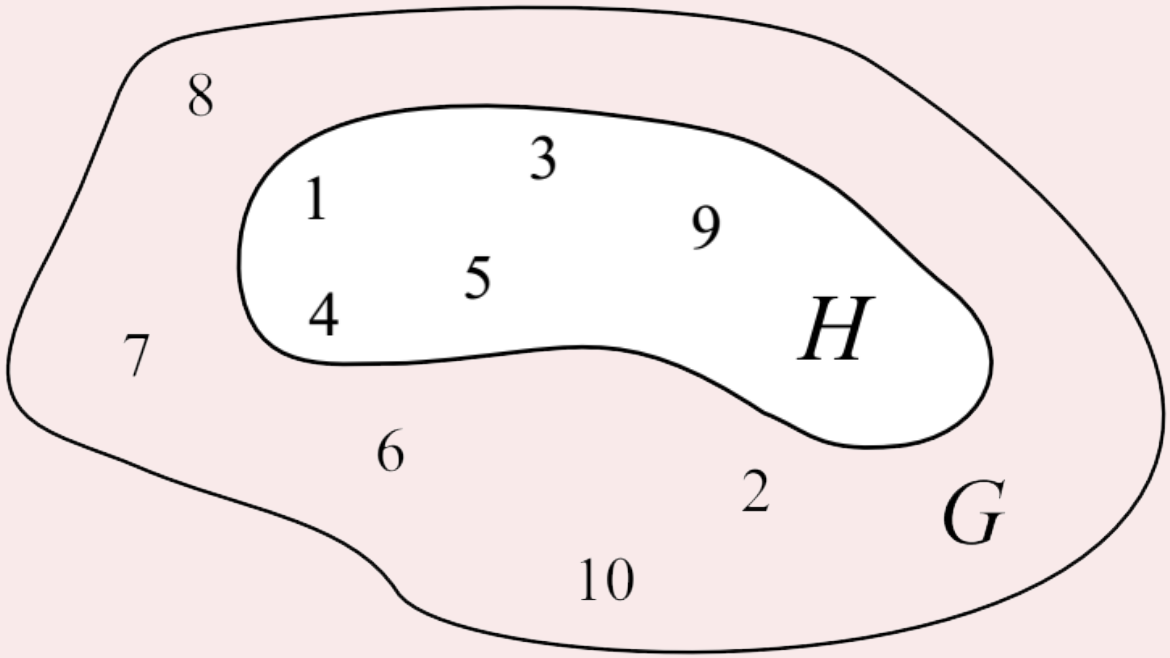
$$\boxed{2,\ 6,\ 7,\ 8}$$

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Subgroups

Subgroups

Definition



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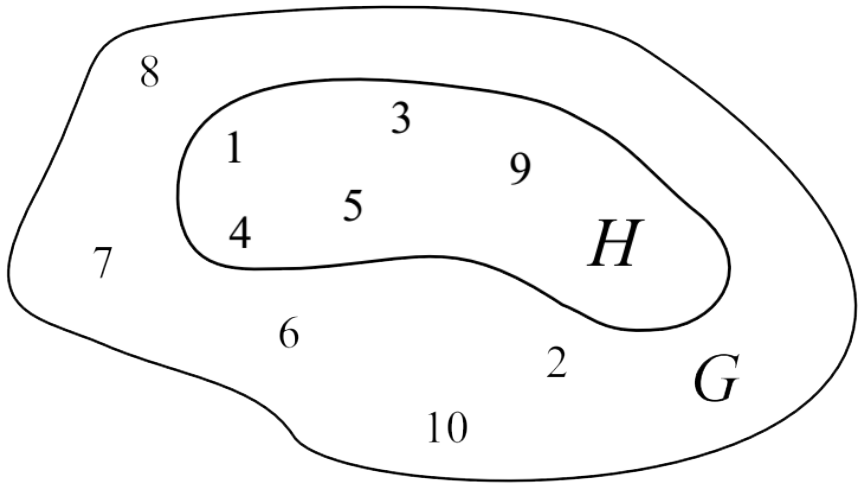
Subgroups

Definition

A **subgroup** is a subset of a group that is also a group under the same operation.

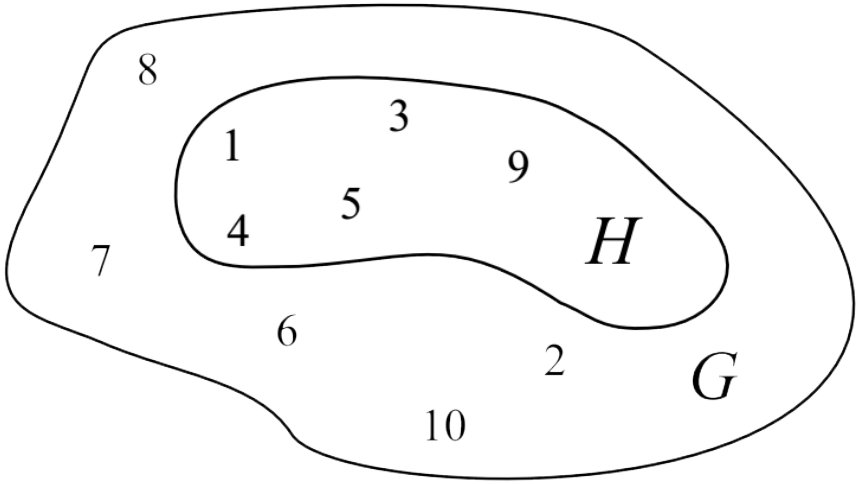
$$\boxed{H \leq G}$$

means that H is a subgroup of G.



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Subgroups



Definition (cont.)

For H to be a subgroup of G: $H \subseteq G$
and the elements of H must satisfy the group properties.

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Subgroups

Cyclic Subgroup Theorem

Let $(G, \circ)$ be a group and let $a \in G$.
If

$$\text{ord}(a) = s,$$

then the powers of a form a cyclic subgroup:

$$\boxed{\langle a \rangle = \{e, a, a^2, \ldots, a^{s-1}\}}$$

and

$$\boxed{|\langle a \rangle| = s}$$

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Example: A Subgroup of $\mathbb{Z}_{11}^*$

Consider the group

$$G = \mathbb{Z}_{11}^*.$$

Since

$$\text{ord}(3) = 5,$$

the element 3 generates the cyclic subgroup

$$\boxed{H = \langle 3 \rangle = \{1, 3, 4, 5, 9\}}$$

with $|H| = 5$

The powers of 3 modulo 11 are

$$3^1 \equiv 3, \quad 3^2 \equiv 9, \quad 3^3 \equiv 5, \quad 3^4 \equiv 4, \quad 3^5 \equiv 1 \pmod{11}.$$

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Example: A Subgroup of $\mathbb{Z}_{11}^*$ (cont.)

Thus,

$$\boxed{H \leq \mathbb{Z}_{11}^*}$$

The multiplication table below verifies that the operation remains inside H.

$\times \bmod 11$	**1**	**3**	**4**	**5**	**9**
1	1	3	4	5	9
3	3	9	1	4	5
4	4	1	5	9	3
5	5	4	9	3	1
9	9	5	3	1	4

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Subgroups

Theorem: Groups of Prime Order

Let H be a finite group of prime order p.
Then H is cyclic, and every non-identity element is a generator of H.

$$\boxed{|H| = p \implies \text{ord}(a) = p \quad \forall\, a \in H,\ a \neq e}$$

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A Subgroup of $\mathbb{Z}_{11}^*$.

Consider the subgroup

$$H = \langle 3 \rangle = \{1, 3, 4, 5, 9\} \leq \mathbb{Z}_{11}^*.$$

Since

$$|H| = 5,$$

and 5 is prime, every non-identity element of H has order 5. Therefore,

$$\text{ord}(3) = \text{ord}(4) = \text{ord}(5) = \text{ord}(9) = 5.$$

Hence, all non-identity elements are generators:

$$\boxed{H = \langle 3 \rangle = \langle 4 \rangle = \langle 5 \rangle = \langle 9 \rangle}$$

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Subgroups

Lagrange's Theorem

Let G be a finite group and let H be a subgroup of G. Then the order of H divides the order of G:

$$\boxed{|H| \mid |G|}$$

In other words, there exists an integer k such that

$$|G| = k\,|H|.$$

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Example: Lagrange's Theorem

Consider $G = \mathbb{Z}_{11}^*$, $|G| = 10$.
The element 3 generates the subgroup

$$H = \langle 3 \rangle = \{1, 3, 4, 5, 9\},$$

so $|H| = 5$.

By Lagrange's theorem,

$$|H| \mid |G|.$$

Indeed, $\boxed{5 \mid 10}$
Thus, the order of the subgroup divides the order of the group.

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Subgroups of $\mathbb{Z}_{11}^*$

The cyclic group

$$G = \mathbb{Z}_{11}^*$$

has order

$$|G| = 10.$$

By Lagrange's theorem, the order of every subgroup $H \leq G$ must divide 10:

$$\boxed{|H| \mid 10}$$

Therefore, the possible subgroup orders are

$$1,\ 2,\ 5,\ 10.$$

The proper subgroups have orders 1, 2, and 5.

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Subgroups of $\mathbb{Z}_{11}^*$ (cont.)

Subgroup	Elements	Generators
H_1	$\{1\}$	1
H_2	$\{1, 10\}$	10
H_5	$\{1, 3, 4, 5, 9\}$	$3, 4, 5, 9$

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Subgroups

Theorem: Constructing Subgroups

Let $G = \langle \alpha \rangle$ be a finite cyclic group of order

$$|G| = n.$$

If k divides n,

$$k \mid n,$$

there exists exactly one subgroup of G with order k.
This subgroup is

$$\boxed{H_k = \left\langle \alpha^{n/k} \right\rangle}$$

and

$$\boxed{|H_k| = k}.$$

Moreover,

$$H_k = \{\, a \in G \mid a^k = e \,\}.$$

Every subgroup of a finite cyclic group is obtained in this way.

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Example: A Subgroup of $\mathbb{Z}_{11}^*$

Consider

$$G = \mathbb{Z}_{11}^*$$

with generator

$$\alpha = 2$$

and

$$|G| = 10.$$

Suppose we want a subgroup of order

$$k = 5.$$

Since

$$5 \mid 10,$$

the theorem tells us to compute

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Example: A Subgroup of $\mathbb{Z}_{11}^*$ (cont.)

$$\alpha^{n/k} = 2^{10/5} = 2^2 = 4.$$

Therefore,

$$H = \langle 4 \rangle.$$

Now compute the powers of 4 modulo 11:

$$4,\ 5,\ 9,\ 3,\ 1.$$

Thus,

$$\boxed{H = \{1, 3, 4, 5, 9\}}$$

and

$$\boxed{|H| = 5}.$$

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Exercise: Construct a Subgroup of Order 2

Consider the cyclic group

$$G = \mathbb{Z}_{11}^*, \qquad |G| = 10.$$

Let $\alpha = 8$ be a generator of G.

Construct a subgroup H of order $k = 2$.
Use the theorem

$$\boxed{\beta = \alpha^{n/k}}$$

where $n = |G|$.

Tasks:

1. Compute the generator β.
2. Find the elements generated by β.
3. Verify that $|H| = 2$.

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Demonstration

We have

$$n = 10, \qquad k = 2, \qquad \alpha = 8.$$

The subgroup generator is

$$\beta = \alpha^{n/k} = 8^{10/2} = 8^5.$$

Modulo 11,

$$8^5 \equiv 10 \pmod{11}.$$

Therefore,

$$\boxed{\beta = 10}.$$

Now compute its powers:

$$10^1 \equiv 10 \pmod{11}$$

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Demonstration (cont.)

$$10^2 \equiv 1 \pmod{11}.$$

Thus,

$$\boxed{H = \langle 10 \rangle = \{1, 10\}}$$

and

$$\boxed{|H| = 2}.$$

Therefore, 10 generates the subgroup of order 2.

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Cryptography

Diffie-Hellman Key Exchange and the Discrete Log Problem

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September 28, 2026

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